Paul Yoon

+1 (713) 320-1032 | pauljy@stanford.edu | linkedin.com/in/pauljinyoon | paulyoon.xyz

EDUCATION

STANFORD UNIVERSITY Palo Alto, CA

Bachelor of Science, Computer Science; Minor in Music | GPA: 3.8/4.0 *Expected June 2027*

Relevant Coursework: Linear Algebra, Differential, and Integral Calculus of Several Variables, Computer Organization and Systems, Data Structures and Algorithms, Probability Theory, Machine Learning, Deep Learning for Computer Vision

PROFESSIONAL EXPERIENCE

AMAZON WEB SERVICES (AWS) Santa Clara, CA

Incoming Software Development Engineer Intern *Jun 2026 – Aug 2026*

- Summer 2026: Sagemaker Hyperpod Team

STANFORD CENTER FOR ARTIFICIAL INTELLIGENCE IN MEDICINE & IMAGING Palo Alto, CA

Researcher – Principal Investigator: Bao Do, MD *Sept 2025 – Mar 2026*

- Built an automated retrieval pipeline linking radiology reports to case images by orchestrating 5+ external APIs (Gemini, Google Images, and internal libraries), removing ∼90% of manual lookup time for clinicians
- Implemented structured-text extraction logic to convert unformatted radiology prose into typed clinical features (location, borders, signal characteristics, enhancement), enabling consistent downstream reasoning and retrieval
- Developed fault-tolerant request sequencing, retries, and response validation to handle high-variance API formats and ensure consistent output quality

SUNDIAL Palo Alto, CA

Software Engineering Intern *Jul 2024 – Sep 2024*

- Implemented a modular Python forecasting component using Facebook Prophet, outperforming the existing production model by 120% (MAPE) and enabling drop-in replacement for the company's analytics pipeline
- Designed a lightweight anomaly detector that reduced false positives for power users, using Spark SQL on S3-backed event logs to compute behavior-based thresholds
- Delivered clean, well-documented model code adopted directly into Sundial's production analytics pipeline

PROJECTS

Hidden Studios AdTech Platform

Next.js, Supabase, TypeScript *May 2025 – Jul 2025*

- Built a backend data model for campaign scheduling, including tables for campaigns, ads, time slots, and linked media
- Implemented ad impression and forecast logic by scraping Fortnite.gg player-count data through custom API endpoints
- Decomposed a 1500+ line Next.js dashboard into modular components, fixing critical workflow bugs (incorrect game ID mapping, scheduler desync, forecast parameter issues) for reliable end-to-end campaign creation

Timestamping Video Game Eliminations with Computer Vision

Python, LaTeX *Apr 2025 – Jun 2025*

- Curated a custom detection dataset (165 events, ∼1,650 frames) with color-jitter augmentation strategies
- Fine-tuned YOLOv8-nano (2.5M params) on <200 images, achieving 0.61 F1 and 0.49s mean temporal error, doubling precision compared to a baseline template matcher
- Engineered a lightweight inference pipeline (OpenCV + ffmpeg) that processes a 14-minute VOD in 1.5 minutes on CPU, generating highlight clips with sub-frame accuracy

Explicit/Implicit Heap Allocator

Unix, C *May 2024 – Jun 2024*

- Implemented the "malloc", "realloc", and "free" functions optimizing for request throughput and memory utilization
- Incorporated an explicit list of nodes to assign optimal locations for new memory requests and lower memory fragmentation
- Achieved 91% memory utilization via testing on heap activity memory requests from Emacs, Cmake, and Firefox

TECHNICAL SKILLS

Languages: Python, TypeScript, C++, SQL, C, JavaScript, HTML/CSS

Frameworks/Libraries: React.js, Next.js, React Native, Supabase, Flask, Pandas, NumPy, Scikit-learn, PyTorch, Matplotlib

Developer Tools: Git, Github, Unix/Linux, Vim, VS Code, Apache Spark, Snowflake, Jupyter Notebook